Towards a Cartography of Datacentering

A Multimodal Research Agenda for Innovation Studies

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The Empirical Moment

Metric	Value
Hyperscaler capex 2026	**$750B**
DC electricity 2025	485 TWh (+17%)
AI-specific growth	+50% YoY
Hyperscaler share 2031	67% of all DC

CreditSights Feb 2026; IEA Energy & AI April 2026;
Synergy Research Group

"The largest concentrated infrastructure build of our lifetimes — happening now."

Not a future scenario.
Not a policy proposal.
An arena in formation.

Why Now? The Contested Arena

- **Stargate:** $500B / ~7 GW committed; fragmented (OpenAI, Oracle, SoftBank)
- **PJM electricity:** capacity prices +833% in one year
- **Maine:** statewide moratorium, April 2026
- **Loudoun County:** eliminates by-right DC, March 2025
- **DeepSeek V4-Pro:** re-opens Jevons paradox debate

37,811 DCD articles

scraped April 2026

headline + intro + image URL

2007 – present
full trade press record

The arena is visible in the trade press before it appears in the academic literature.

The Gap in Innovation Studies

RP canon covers:

Platforms (Gawer 2014)
GPTs and productivity (Brynjolfsson)
Sectoral systems (Malerba)
Transitions (Geels)

Critical DC studies exists:

Big Data & Society, *Convergence*
(Edwards et al. 2025; Monstadt 2025)
— but outside RP's intellectual canon

Missing:

Integrative framework for the *infrastructural + methodological*
turn in AI-era innovation studies

No unified approach to:

mapping contested arenas
surfacing implicated actors
operationalizing multimodal evidence

Theory Layer 1: Arenas of Development

Jørgensen (2012) — *Research Policy*

"Mapping and navigating transitions: The MLP compared with arenas of development"

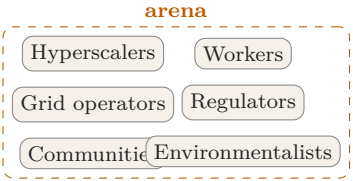
Key move: flat ontology

Start from *actor constellations* and their collective engagement, not pre-classified structural tiers.

Negotiation is the fundamental process.

Allows political, social, environmental spheres simultaneously.

For datacentering:



Co-constitutive — not hierarchically ordered.

Theory Layer 2: Situational Analysis

Clarke (2005) — STS / Grounded Theory

Extends Strauss with three kinds of maps:

1. Situational maps (all human/nonhuman elements)
2. Social worlds / arenas maps
3. Positional maps

Key innovation: implicated actors

(1) Silenced / made invisible by power
(2) Discursively constructed but absent

For datacentering:

Decided by:	boards, planners
Silenced:	communities, visa workers
Invoked:	"the consumer" "future generations"

Who is excluded from the negotiation?
Who is invoked but absent?

Theory Layer 3: Issue Mapping + CITC

Marres (2015)

Issue mapping via digital traces:
hyperlinks, hashtags, discourse networks.
Affirmative approach: traces as
empirical window into issue formation.

Berente, Seidel & Safadi (2019)
Miranda et al. (2022) — CITC

Computationally Intensive Theory Construction:
sampling → synchronic → diachronic
→ **lexical framing**

Patterns become contributions *only*
once situated in scholarly discourse.

Marres tells us
what to trace

CITC tells us
how to trace it
rigorously enough
to make a
theoretical contribution

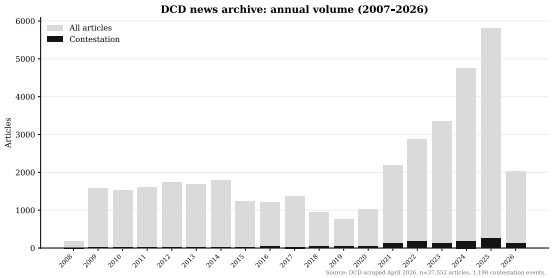
Prevents atheoretical
pattern-mining.
The test for RP.

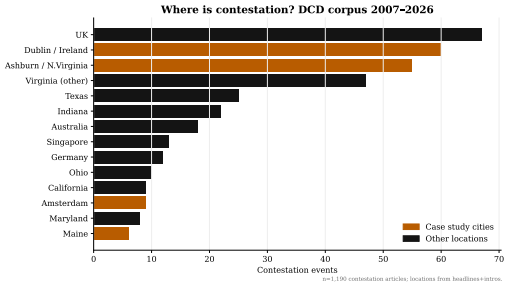
Operationalizing: Multimodal Data Sources

Modality	Source / method	Output
Text	DCD news corpus (37,552 articles, 2008–2026); 1,190 contestation events; topic modelling; registerial analysis (Matthiessen 2015) + GLiNER NER · BERTopic · CITC cycle	Topic distributions + verb process types
Image	Satellite (Sentinel-2, Landsat LST); DCD article images; corporate photography	Land conversion maps; visual register index (CLIP + BERTopic)
Graph	Hyperlink networks; ownership (SEC EDGAR, REIT filings); urban embeddedness (city2graph + Overture Maps)	Arena actor maps; governance counterparty analysis
Structured	Crunchbase / 10-Ks; planning records	Funding flows; capex + geography diffusion

Each modality runs through the CITC cycle. This is operationalized, not theoretical.

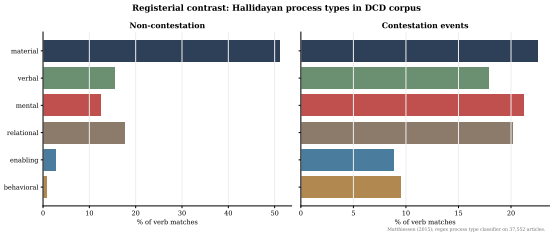
The Arena in Data: 37,552 Articles, 1,190 Contestation Events





Peak contestation year: 2025. Top triggers: legal/court, planning refusal, community opposition, moratorium. **Analysis on headlines + intro text only** (first 1–2 sentences). Full article text via screenshot/OCR pipeline: more coming.

What the Verb Register Shows



Key shift (Matthiessen 2015)

Process type	Contest	Non
Behavioral	23.1%	1.6%
Enabling	16.5%	3.1%
Material	33.7%	66.8%

Doing → enabling/exploring

The arena shifts from building to governing and contesting.

This is the governance gap in its linguistic signature.

Headlines + intros only.

Research Agenda 1: Regional Innovation Systems

Central question: How do hyperscale data-center clusters reshape local productivity, labour markets, energy regimes, and property relations?

Cases: Loudoun (US), Luleå (SE)
Dublin (IE), Hyderabad (IN)

Metrics: Employment volatility
Electricity price shocks
Tax dependence
Community displacement

From DCD corpus:

News framing shift:
pro-growth → environmental concern

City2graph:
residential adjacency + power density

Amsterdam: moratorium <1yr after signal onset.
Virginia: comparable signal, no response.

Research Agenda 2: Sectoral Transformation

Central question: How does hyperscaler vertical integration reshape sectoral innovation systems, especially in AI-using industries?

Mechanism:
AWS, Google, Meta, Microsoft
control compute + models + stacks

Effect:
Changed startup access
New forms of lock-in
Shifted innovation incentives

Evidence:
Hyperlink networks of startup funding
(hyperscaler-mediated vs. traditional VC)

Crunchbase exit patterns

News coverage of
"platform dependency" clusters

Research Agenda 3: Policy Sovereignty & Energy

Central question: What policy mix sustains sovereign compute capacity without fossil-fuel dependence or geopolitical lock-in?

Cases:
UK AI Growth Zones
EU AI Factories
France €109B investment
Saudi HUMAIN · UAE Stargate

Instruments:
Grid access, renewables mandates, labour standards, tax incentives

Evidence:
Policy document discourse SNA
Capital flows to sovereign projects
Regulatory filing timelines
Policy innovation diffusion curves

What This Enables: Methodological Innovation

- **Empirical cartography:** contested infrastructure transitions made visually traceable and empirically accountable
- **Multi-arena mapping:** surfaces who is included, excluded, silenced, or discursively constructed
- **Toolkit for RP:** replicable for energy, telecom, rare earths, supply chains — any contested infrastructure transition
- **Bridge STS ↔ Innovation:** critical infrastructure scholarship → RP policy canon
- **Open science:** methodological infrastructure for multimodal evidence synthesis across disciplines

`github.com/babajideowoyele/datacentering-cartography`

What would innovation studies look like if we treated data infrastructure buildout — both as physical facilities and as research infrastructure — as

foundational rather than peripheral?

A cartography of datacentering is one answer.

What are others?

Backup: Loudoun County Arena (Situational Map)

Actors

Hyperscalers	Amazon AWS, Google, Meta
Energy	Dominion Energy
Planning	Loudoun County Board
Opposition	Sierra Club, Piedmont EC
Labour	visa workers + local

Nonhuman actants

Electricity grid · hyperscale facilities
Zoning laws · satellite time-series

Discourses in corpus

"Tech economic engine"
"Environmental burden"
"Worker precarity"

Registerial signal

Verb register 2019–2024:
material ↑ enabling ↑
relational ↓

Action displacing description
= arena in contestation

Q: Isn't this just STS dressed up as innovation studies?

Live: "Fair concern — I'm using Jørgensen and Marres specifically because they bridge to RP and *ST&HV*, not importing critical theory wholesale."

The methodological spine:
CITC (Berente, Seidel & Safadi 2019; Miranda et al. 2022) comes from *MIS Quarterly* and *ISR* — not from STS.

The agenda is testable, not hermeneutic.

Slide 13 makes the RP positioning explicit.

Q: Where's the empirical contribution?

Live: "The ECR paper is the agenda. The Loudoun pilot is the first empirical instantiation — I can speak to that."

Deliberate sequencing: the agenda paper exists to make the methodology citable *before* the empirical papers, so reviewers of those papers have a coherent framework to reference.

From 37,811 DCD articles:

Analysis	Status	Finding
EDA geography	Complete	US/Europe split visible at N=37K
Register analysis	Running	Verb types by city
BERTopic	Queued	Topic clusters on full corpus
city2graph	Seed run	3 cities, scales with NER

Q: Why CITC and not computational social science more broadly?

Live: "Because CITC's lexical-framing requirement is what makes the output theoretical, not just descriptive. That's the test for RP."

CITC explicitly distinguishes patterns from contributions.

That's the discipline that protects against the most common reviewer objection to digital-trace work:

> *"This is interesting, but what's the theoretical contribution?"*

Q: Why Loudoun, Luleå, Dublin, Hyderabad?

Variation on three dimensions:

Case	Regulatory regime	Energy mix	Industrial trajectory
Loudoun	Common law / NIMBY	Gas + solar	Mature US cluster
Luleå	Swedish planning	98% hydro	Renewable-led
Dublin	Irish planning	Mixed, constrained	FDI platform
Hyderabad	Indian regulatory	Mixed, expanding	Emerging South Asian hub